



TECH-TRIATHLON 2026 | DESIGNATHON

Relay

Connected delivery operations for Waypoint Group

TEAM

Zephyrus

ROLES

Store manager, Dispatcher, Loader, Driver

DOCUMENT

Design rationale and prototype scope

PRIMARY FLOW

Order → Plan → Load → Deliver → Confirm receipt

THE OPERATING PROBLEM

Shared capacity, disconnected decisions.

Waypoint Group's three brands use one delivery network, but they place different demands on it.

Waypoint Fresh

Fresh stores need deliveries before opening, and chilled or frozen goods require refrigerated vehicles. Some outlets are van-only.

Waypoint Style

Garment loads often reach their volume limit before their weight limit. Mall stores also have fixed access windows.

Waypoint Tech

Appliances and electronics are heavy, fragile, and valuable. Daily demand can vary.

Dispatchers currently plan with spreadsheets and operational knowledge. Instructions move through calls, dock conversations, printed run sheets, and handwritten notes. This makes several handoffs difficult to keep current:

- Orders can be re-entered when they move from a call or message into the plan.
- Loading lists can become outdated when a plan changes.
- Dispatch has no shared view of route progress after departure.
- Deferral decisions can lack a clear record of reason and consequence.
- Proof of delivery can remain fragmented across memory, paper, and calls.
- Field connectivity can fail while drivers still need to record work.
- Future demand is difficult to turn into a capacity conversation.

Order → **Plan** → **Load** → **Deliver** → **Confirm receipt**

DESIGN OBJECTIVE AND SCOPE

One connected record across four working contexts.

Relay connects ordering, planning, loading, delivery, and receipt so each role can act on the previous role's work.

STORE MANAGER**Order and receive**

Choose full-carton replenishment, submit before the 16:00 cutoff, track the planned arrival, inspect driver POD, confirm receipt, and report an issue.

DISPATCHER**Plan and explain**

Work from one confirmed queue, assign a route and vehicle, validate constraints, record deferrals, monitor delivery, and review receipt follow-up.

LOADER**Prepare the route**

Use the released stop sequence, load the final stop first, check cartons and products, and resolve short or damaged shipments before departure.

DRIVER**Deliver and prove**

See one next stop on a phone, record arrival, report exceptions, return to deferred stops, and capture carton verification, recipient, and delivery time.

High-fidelity slice

The implemented walkthrough follows one complete Waypoint Fresh trip, R-07 in Colombo, from Peliyagoda across all four roles. Style and Tech appear as scheduled previews. This keeps the critical handoffs testable without claiming that every route is editable.

Included

- Four fixed-role demo accounts
- Shared order lifecycle
- Planning validation and deferral record
- Reverse loading and issue resolution
- Delivery exceptions and receipt capture
- Simulated offline queue and recovery

Deliberately bounded

- No public signup or editable roles
- No server-side authorization
- No live maps, contacts, or cross-device state
- No real allocation optimizer
- No submitted forecasting model
- No actual service worker synchronization

OPERATING CONSTRAINTS

The plan is explainable because its rules are visible.

Constraint	Why it matters	How Relay surfaces it
Weight	Every vehicle has a payload limit.	Planned kilograms are shown against vehicle capacity. Overweight assignments are blocked.
Volume	A load must fit by volume as well as weight.	Planned cubic meters are shown beside the vehicle's volume capacity.
Refrigeration	Chilled and frozen goods require a refrigerated vehicle.	Order temperature and vehicle capability appear in the queue, fleet choice, and validation checks.
Van-only access	Some outlets cannot be served by trucks.	The outlet access condition stays visible, and an incompatible assignment explains why it is blocked.
Depot	Orders and vehicles must align with the serving depot.	The route, order, and selected vehicle identify Peliyagoda and validate depot compatibility.
Fuel quota	Route distance consumes the vehicle's weekly allowance.	Estimated route distance, vehicle fuel use, and weekly quota appear in planning checks.
Trips and time	A vehicle may run up to two routes per day, and the Fresh run has a practical time budget.	Trip 1 of 2 and route minutes are visible before release.
Delivery windows	Fresh stores must receive before opening, while mall access is fixed.	Projected arrival is checked against each order window and at-risk stops are identified.

Order timing

Next-day orders close at 16:00. The Store workspace shows the cutoff, and the Dispatch workspace identifies the operating day and next-day plan.

CONNECTED WORKFLOW

One order becomes the next role's work.

Store manager: order

Chooses full cartons and confirms replenishment before the cutoff.

Dispatcher: plan and allocate

Assigns route and vehicle, validates constraints, and releases the plan.

Loader: load

Checks shipments and loads the final stop first.

Driver: deliver

Records arrival, exceptions, recipient, cartons, and time.

Store manager: confirm receipt

Inspects the POD, confirms cartons, or reports an issue.

The screenshot displays the Relay planning workspace, titled "Next-day dispatch" for the location "Colombo". It shows a route map with stops and a detailed list of orders. The interface includes a sidebar with navigation options (Dispatch, Account), a main workspace with a route map and order details, and a right sidebar with vehicle information and a "Confirm plan" button. The route map shows a path starting from a depot, visiting several stops, and returning. The order details list includes columns for order ID, location, quantity, and status. The right sidebar shows vehicle information (VEH003) and a "Confirm plan" button.

Relay planning workspace showing route allocation and feasibility validation.

Dispatcher | Plan & validate

The screenshot is captured from the running Relay prototype at a desktop viewport. The route map is schematic and is not used for navigation.

KEY DESIGN DECISIONS

Each decision protects a real handoff.

1. Assisted planning with visible validation

Reason: The dispatcher holds operating knowledge and remains accountable for tradeoffs.

Prototype: Seven validation groups appear beside the stop sequence and vehicle.

2. Explainable deferrals

Reason: An unserved outlet needs a recorded reason and next action, especially after a previous skip.

Prototype: The deferral requires both a reason and impact note, then becomes visible to Store.

3. Offline field work

Reason: Coverage can drop away from the depot, but a driver or loader still needs to record work.

Prototype: Safe actions persist locally, pending updates remain visible, and recovery clears the simulated queue.

4. Reverse loading sequence

Reason: The first delivery needs to be nearest the vehicle door.

Prototype: The manifest starts with the final route stop and blocks out-of-sequence confirmation.

5. Shared order lifecycle

Reason: Separate role dashboards would repeat the same fragmented handoff.

Prototype: Order status, route, loading progress, delivery, and receipt use one shared browser record.

6. Explicit proof of delivery

Reason: Delivery disputes should not depend on memory or an incomplete note.

Prototype: Full carton count, verification, recipient, and time are required before delivery completes.

7. Future capacity outlook

Reason: Demand planning needs a discussion about vehicles, drivers, and refrigerated capacity.

Prototype: A clearly labeled illustrative 10-week view demonstrates that decision without claiming real forecasting.

PROTOTYPE BOUNDARY

What Relay demonstrates, and what remains simulated.

Demonstrated in the prototype

- Responsive interfaces for all four roles
- One connected order, plan, load, delivery, and receipt flow
- Constraint visibility and blocked incompatible assignments
- Reasoned deferral with Store visibility
- Reverse loading and issue resolution
- Delivery exceptions, return visits, and POD validation
- Store receipt confirmation and issue reporting
- Local persistence through refresh
- Simulated offline status, queued updates, and recovery feedback

Simulated or illustrative

- Operational records, stock cover, suggestions, routes, and ETAs
- Contacts and navigation to an external map search
- Forecast values and capacity recommendations
- Authentication and authorization in the frontend
- Synchronization through a local pending-update array
- All persistence in the browser's relay-v1 localStorage entry

Not claimed

Relay has no backend, database, real account provisioning, production authorization, live cross-device synchronization, service worker, route optimizer, submitted forecasting model, or production infrastructure.

R-07 Fresh is the one editable end-to-end trip. R-12 Style and R-15 Tech are scope previews. Partial deliveries remain outstanding for reconciliation, and a driver POD does not update Store stock until the Store manager confirms receipt.

AI TOOL DISCLOSURE

How AI assisted this submission.

AI-assisted

Requirements extraction, gap analysis, source-alignment suggestions, front-end code revision, test expansion, copy refinement, and beginner documentation were assisted by OpenAI Codex during the final review.

Human-led and existing work

The original Relay concept, visual direction, interaction model, and submitted workspace were provided by the team. Final competition claims, design judgment, prototype demonstration, and submission decisions remain the team's responsibility.

Verification

Official booklet requirements and supplied CSV schemas were checked directly. Automated browser tests exercise the connected workflow, exceptions, offline recovery, accessibility behaviors, and responsive layouts.

CORE TRADEOFF

Explainability before invisible automation.

Relay keeps the dispatcher in control and makes feasibility checks visible. The system blocks an infeasible choice, but it does not pretend to know the only correct allocation.

What this protects

Operational judgment, accountability for deferrals, and a plan that can be explained to Stores and field teams.

What this gives up

The prototype does not provide automatic route optimization or claim that its illustrative plan is optimal.

Prepared from the implemented Relay application, its README and case-study source, and the official Tech-Triathlon 2026 Challenge Booklet.